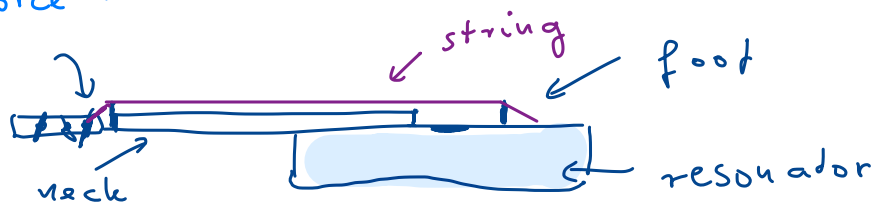
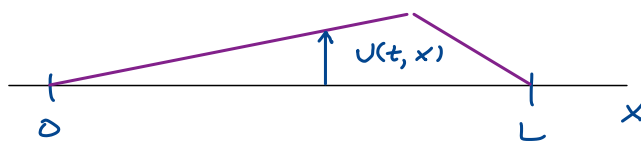


# Partial differential equations

Consider the problem of a string  
force  $F$

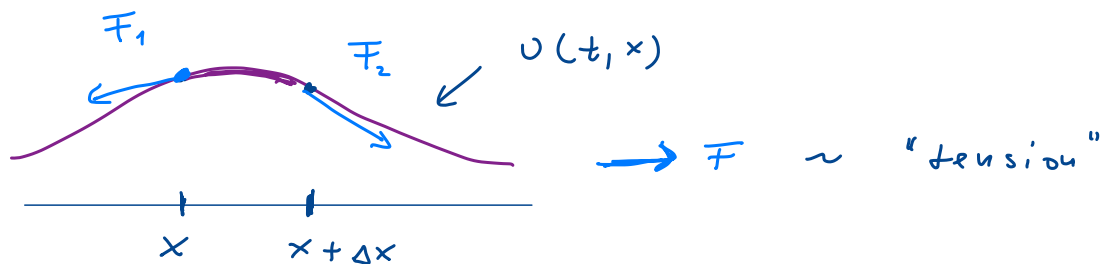


$t=0$



at  $t=0$   $U(0, x) = f(x) \Rightarrow Q: U(t, x) = ?$

first construct equation of motion



vertical component of force

$$\begin{aligned} F_1^v &= -F \frac{\partial U}{\partial x}(x) + F \frac{\partial U}{\partial x}(x+\Delta x) \approx \\ &\approx F \cdot \Delta x \cdot \frac{\partial^2 U}{\partial x^2}(x) \end{aligned}$$

Newton:  $m = \underset{\substack{\uparrow \\ \text{linear mass density}}}{\rho} \cdot \Delta x$   $[\rho] = \left[ \frac{\text{kg}}{\text{m}} \right]$

$a = \frac{\partial^2 U}{\partial t^2}(x)$  acceleration

$$\Rightarrow \rho \cdot \Delta x \cdot \frac{\partial^2}{\partial t^2} u(t, x) = \Delta x \cdot F \cdot \frac{\partial^2}{\partial x^2} u(t, x)$$

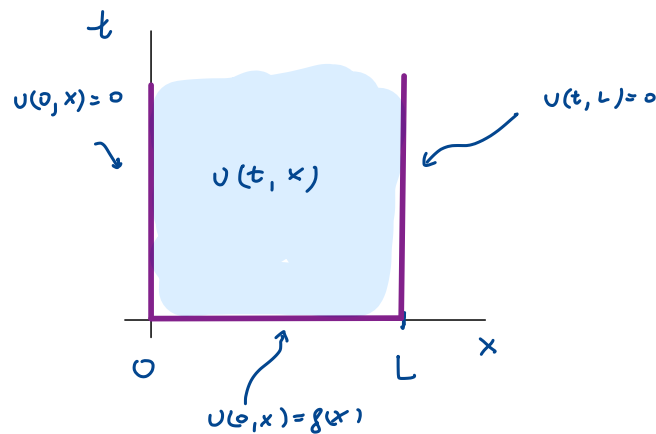
$$\frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$$

1D  
wave  
equation

$$c^2 = \frac{F}{\rho}, \quad [c^2] = \frac{[F]}{[\rho]} = \frac{\left[ \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \right]}{\left[ \text{kg/m} \right]} = \left[ \frac{\text{m}^2}{\text{s}^2} \right]$$

$\Rightarrow c$  : sound velocity

Boundary condition



Def. partial differential equation = differential equation of a multivariable function, relating partial derivatives

Examples

- $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

Laplace eqn.  
in 2D

- $\frac{\partial}{\partial t} g(t, x) = D \cdot \frac{\partial^2 g(t, x)}{\partial x^2}$

1D diffusion  
eqn.

- $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = -\frac{1}{\epsilon_0} g(x, y, z)$

3D Poisson

# Solution of string problem

$$(1) \quad \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0$$

boundary conditions:

$$u(0, x) = f(x), \quad \frac{\partial u}{\partial t}(0, x) = g(x)$$

$$u(t, 0) = u(t, L) = 0.$$

## Separation of variables

Ansatz:  $u(t, x) = ? \quad F(x) \cdot G(t)$

Q: are there solutions of this form?  
 $\Rightarrow$  use these to construct solutions

$$(1) \Rightarrow \frac{1}{c^2} G''(t) \cdot F(x) - G(t) F''(x) = 0$$

$$\Rightarrow \frac{1}{c^2} \frac{G''(t)}{G(t)} = \frac{F''(x)}{F(x)}$$

can only be equal if both sides are constant

$$\frac{1}{c^2} \frac{G''(t)}{G(t)} = \frac{F''(x)}{F(x)} = \kappa \leftarrow \text{parametrizes solution relates } F(x) \text{ and } G(t)$$

$$(2) \quad \frac{1}{c^2} G''(t) = \kappa \cdot G(t)$$

$$F''(x) = \kappa \cdot F(x) \quad (2')$$

2 ODEs!  
 $\uparrow$   
connected by  $\kappa$ !

• if  $\kappa = 0 \Rightarrow F'' = 0 \Rightarrow F(x) = a \cdot x + b$

boundary condition: if  $G(t) \neq 0$  at any time

$$\Rightarrow F(0) = F(L) = 0$$

can only be satisfied by  $a = b = 0$   $\hookrightarrow$

• if  $\kappa > 0 \Rightarrow F'' = \kappa F \quad F_{\pm}(x) = e^{\pm \sqrt{\kappa} \cdot x}$

$$F(x) = a \cdot e^{\sqrt{\kappa} \cdot x} + b \cdot e^{-\sqrt{\kappa} \cdot x}$$

$$a + b = 0$$

$$a \cdot e^{\sqrt{\kappa} \cdot L} + b \cdot e^{-\sqrt{\kappa} \cdot L} = 0$$

$$\Rightarrow a = b = 0 \quad \hookrightarrow$$

$$\kappa < 0 \Rightarrow \kappa = (i\lambda)^2 = -\lambda^2$$

$$\Rightarrow \text{solutions } a \cdot e^{i\lambda \cdot x} + b \cdot e^{-i\lambda \cdot x} = F(x)$$

$$F(0) = 0 \Rightarrow a = -b$$

$$F(L) = 0 = a(e^{i\lambda \cdot L} - e^{-i\lambda \cdot L}) \Rightarrow \lambda_n = \frac{\pi}{L} \cdot n$$

$$a \rightarrow \frac{1}{2i} \quad e^{i\lambda \cdot 2L} = 1 \quad n = 1, \dots$$

$$\Rightarrow F_n(x) = \sin(\lambda_n \cdot x), \quad \lambda_n = n \cdot \frac{\pi}{L} \Rightarrow K_n = -\frac{n^2 \pi^2}{L^2}$$

solve for  $G(t)$

$$\frac{1}{c^2} G_n''(t) = -\lambda_n^2 G_n(t) \Rightarrow G_n(t) = e^{\pm i\lambda_n \cdot c \cdot t}$$

$$\text{or } \Rightarrow G_n(t) = \begin{cases} \sin(\lambda_n \cdot c \cdot t) \\ \cos(\lambda_n \cdot c \cdot t) \end{cases}$$

$\Rightarrow$  solutions are of form

$$\cos(\lambda_n \cdot c \cdot t) \sin(\lambda_n \cdot x), \quad \sin(\lambda_n \cdot c \cdot t) \sin(\lambda_n \cdot x) \quad n = 1, 2, \dots$$

(1) is a homogeneous, linear equation!

$\Rightarrow$  linear combinations of solutions are solutions, too!

$\Rightarrow$  general solution:

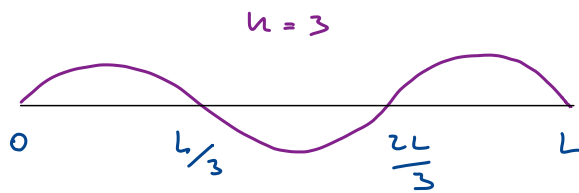
$$u(x, t) = \sum_{n=1}^{\infty} \left( A_n \cdot \cos(\lambda_n \cdot c \cdot t) + B_n \sin(\lambda_n \cdot c \cdot t) \right) \cdot \sin(\lambda_n \cdot x)$$

$$\lambda_n = \frac{\pi}{L} \cdot n$$

eigenmode

consider  $\sin(\lambda_n \cdot x) = \sin\left(n \cdot \frac{\pi}{L} \cdot x\right)$

has nodes at  $\frac{L}{n}, \frac{2L}{n}, \dots, \frac{(n-1) \cdot L}{n}$  —  $n-1$  nodes



eigen frequency :

$$\omega_n = c \cdot \lambda_n = n \frac{c \cdot \pi}{L}$$

remark :  $c = \sqrt{\frac{F}{\rho}}$  ← force

$$\omega_1 = \sqrt{\frac{F}{\rho}} \cdot \frac{\pi}{L}$$

↑  
tuning !

Boundary condition :

$$t=0 : \quad U(0, x) = \sum_{n=1}^{\infty} \alpha_n \cdot \sin\left(n \cdot \frac{\pi}{L} \cdot x\right) \stackrel{!}{=} f(x)$$

$$\Rightarrow \alpha_n = \frac{2}{L} \int_0^L dx \sin\left(n \cdot \frac{\pi}{L} \cdot x\right) \cdot f(x)$$

$$\partial_t U(0, x) = \sum_{n=1}^{\infty} \beta_n \cdot n \cdot \frac{\pi}{L} \cdot \sin\left(n \cdot \frac{\pi}{L} \cdot x\right) \stackrel{!}{=} g(x)$$

$$\Rightarrow n \cdot \frac{\pi}{L} \cdot \beta_n = \frac{2}{L} \int_0^L dx \sin\left(n \cdot \frac{\pi}{L} \cdot x\right) \cdot g(x)$$

$$\beta_n = \frac{2}{n \cdot \pi} \cdot \int_0^L dx \sin\left(n \cdot \frac{\pi}{L} \cdot x\right) \cdot g(x)$$

consider simple case:  $\partial_t u(t=0, x) = g(x) \equiv 0$ !

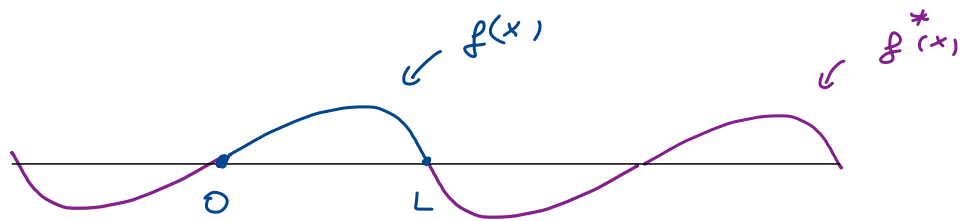
Then 
$$u(x, t) = \sum_{n=1}^{\infty} \alpha_n \cdot \cos\left(n \cdot \frac{\pi}{L} \cdot c \cdot t\right) \cdot \sin\left(n \cdot \frac{\pi}{L} \cdot x\right)$$

use  $\sin(\alpha) \cdot \cos(\beta) = \frac{1}{2} (\sin(\alpha + \beta) + \sin(\alpha - \beta))$

$$\Rightarrow u(x, t) = \frac{1}{2} \sum_{n=1}^{\infty} \alpha_n \left\{ \sin\left(n \cdot \frac{\pi}{L} \cdot (x + ct)\right) + \sin\left(n \cdot \frac{\pi}{L} \cdot (x - ct)\right) \right\}$$

$$= \frac{1}{2} \left( \overset{\substack{\text{left mover} \\ \downarrow}}{f^*(x+ct)} + \overset{\substack{\text{right mover} \\ \nwarrow}}{f^*(x-ct)} \right)$$

↗  
antisymmetrical extension of  $f(x)$



Back to original problem:



$$f(x) = \begin{cases} k \cdot x & x < \frac{L}{2} \\ k \cdot (L-x) & x > \frac{L}{2} \end{cases}$$

$$\alpha_n = \frac{2}{L} \int_0^L dx \sin\left(n \cdot \frac{\pi}{L} \cdot x\right) f(x) \underset{n \text{ odd} = 2\ell+1}{=} \dots$$

$$= k \cdot \frac{(-1)^\ell \cdot 4 \cdot L}{\pi^2 \cdot (2\ell+1)^2}$$

$$\ell = 0, 1, \dots$$

$$u(x, t) = \frac{4k \cdot L}{\pi^2} \left\{ \sin\left(\frac{\pi}{L} \cdot x\right) \cos\left(\frac{\pi}{L} \cdot c \cdot t\right) \right.$$

✓

← octave missing!

$$- \frac{1}{3^2} \sin\left(\frac{3\pi}{L} \cdot x\right) \cdot \cos\left(\frac{3\pi}{L} \cdot c \cdot t\right)$$

← octave + fifth  
2 · 3/2

✓

← (octave)<sup>2</sup> missing  
2 · 2

$$+ \frac{1}{5^2} \sin\left(\frac{5\pi}{L} \cdot x\right) \cdot \cos\left(\frac{5\pi}{L} \cdot c \cdot t\right)$$

2 octave + third  
(large)

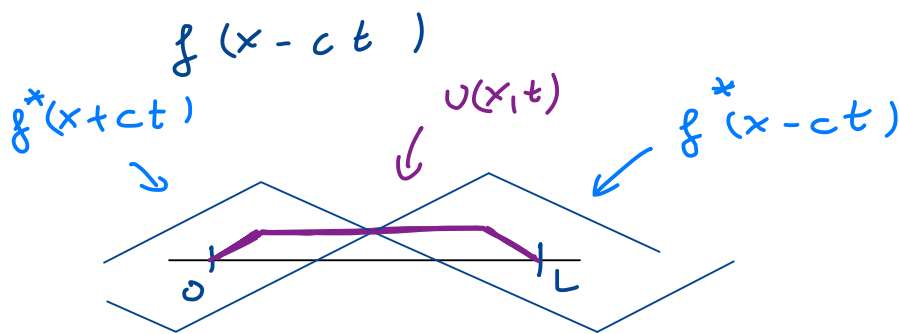
$$2 \cdot 2 \cdot \frac{5}{4}$$

+ .. }  
}

⇒ if you want to get other overtones →  
hit the cord at a side!

for high pitch → edge of cord!

Second approach



d'Alembert's solution:

Statement:  $u(t, x) = \phi(x - c \cdot t) + \psi(x + c \cdot t)$   
 ↑ appropriate functions

$$\text{B.C. (1)} \quad u(0, x) = f(x) = \phi(x) + \psi(x)$$

$$(2) \quad \partial_t u(0, x) = g(x) = c \cdot \psi'(x) - c \cdot \phi'(x)$$

$$\Rightarrow \frac{1}{c} \int_{x_0}^x g(x) + \underbrace{\psi(x_0) - \phi(x_0)}_{= c(x_0)} = \psi(x) - \phi(x)$$

$$\Rightarrow \psi(x) = \frac{1}{2} \left( f(x) + \frac{1}{c} \int_{x_0}^x g(x) + c(x_0) \right)$$

$$\phi(x) = \frac{1}{2} \left( f(x) - \frac{1}{c} \int_{x_0}^x g(x) - c(x_0) \right)$$

$$\Rightarrow u(x, t) = \phi(x - ct) + \psi(x + ct) =$$

$$= \frac{1}{2} \left( f(x - ct) + f(x + ct) \right) + \frac{1}{c} \int_{x - ct}^{x + ct} g(x') dx'$$

$$u(x, t) = \frac{1}{2} \left( f(x - ct) + f(x + ct) \right) + \frac{1}{c} \int_{-ct}^{ct} dx' g(x + x')$$